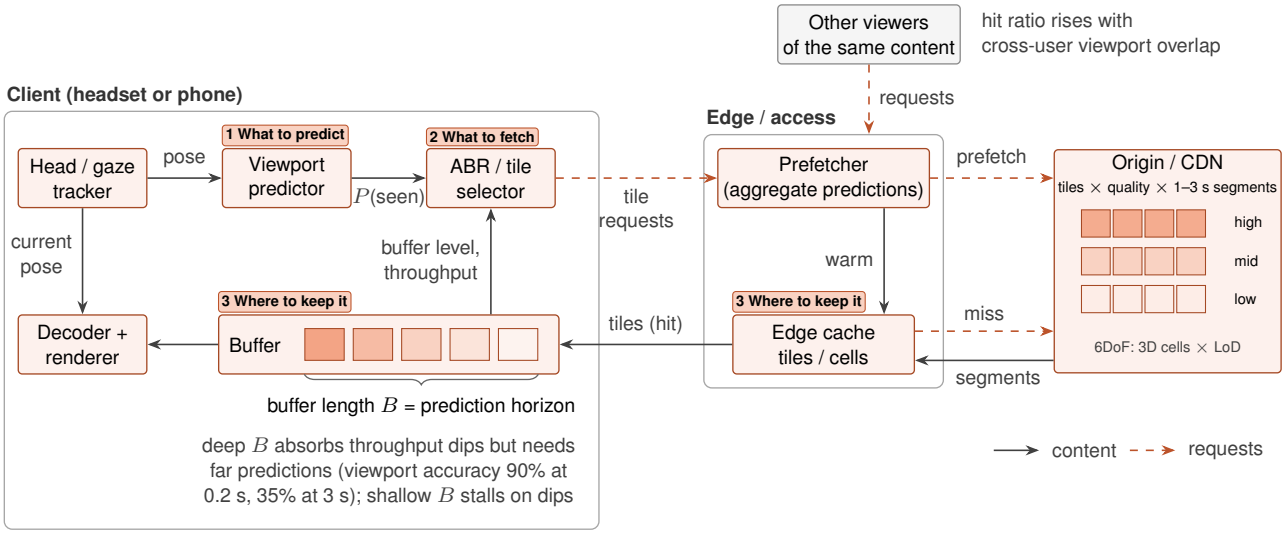


Figure 1: Viewport-adaptive delivery of tiled 360-degree or volumetric content across client, edge and origin, with the three anticipation decisions marked. Each decision is a bet on where the user will look: the buffer length fixes how far ahead the predictor must see, and the edge cache pays off only when viewers' viewports overlap.

1 Delivering Ahead of the User: Prediction, Adaptation and Caching

An immersive client displays only a fraction of what it could receive, and it must request that fraction before the user looks there. Delivery is therefore a bet on the user's future attention. Three decisions place the bet: *what to predict* about the user, *what to fetch* given that prediction and the network, and *where to keep* content so that it is close when needed. They form one loop (Figure 1). The predictor's horizon limits how deep the buffer may be, the buffer's depth fixes how far ahead the fetcher commits bandwidth, and the caches upstream decide how expensive a wrong commitment is. This section treats each decision in turn, summarises the families of approaches in Table 1, and then follows the coupling, which explains why measured savings shrink as evaluations become realistic. How content is tiled and represented is the subject of **??**; the same predictions hide rendering delay in **??**; the radio and transport that carry the fetched bits are treated in **??**.

1.1 What to predict: horizon against accuracy

Prediction trades horizon for accuracy, and the right target depends on what the content lets the user do. For 3DoF video the target is head orientation, from which the viewport follows. For volumetric content it is the full 6DoF pose, because position, distance and occlusion change what is visible. Gaze narrows the target to the foveal region, while content and other viewers supply priors that do not depend on the user's own history. Accuracy falls steeply with horizon. On a 130-user trace set, Flare's best predictor reached median accuracies of 90.5%, 72.9%, 58.2% and 35.2% at 0.2, 0.5, 1 and 3 s [1], and a cross-dataset benchmark found no algorithm accurate much beyond 300 ms [2]. Head motion is predictable over 100–500 ms [3]; streaming buffers want seconds.

Three families of predictors address this gap. *Extrapolators* (persistence, linear regression, Kalman filters) assume that motion stays smooth over the horizon; they need no training and cost almost nothing, and Kalman filters are the usual choice for 6DoF pose in cloud volumetric rendering [4]. *Learned sequence models* (LSTM, GRU, transformer) learn motion patterns from the user's past trajectory; VPT360, for instance, uses only the scanpath [5]. *Content-aware and cross-user predictors* add what a trajectory cannot know: saliency and motion maps of the frame [6], gaze [7], or the paths of earlier viewers. Earlier viewers are the strongest long-horizon signal for on-demand content. Cross-user prediction raised accuracy by 20.2 percentage points [8], clustering users by fixation added 15% precision over linear regression [9], and other users' paths helped most at 4–10 s [10]. Live content has no such history and relies on online learning or object semantics [11, 12]. Interactive XR has neither future frames nor other viewers, so these results do not transfer to it.

Whether learned models beat simple ones is contested. On the Tsinghua dataset, persistence (keep the current viewport) reached a hit rate of 0.8355, against 0.8402 for a sequence-to-sequence model and 0.8464 for an LSTM [10]. For 6DoF pose, simple statistical models performed comparably well

Table 1: Families of approaches for the three anticipation decisions. Each family wins under its own assumption; none dominates once that assumption fails.

Decision	Family	Idea	Assumes	Wins when	Limit
Predict	Extrapolators	Persistence, linear or Kalman continuation of pose	Smooth motion over the horizon	Horizon $<$0.5 s; no training data	Blind to content and to sudden turns
	Learned sequence	LSTM or transformer over past trajectory	Training users and videos resemble test ones	Mid horizons on familiar content	Small margin over persistence; leakage-prone splits
	Content-aware, cross-user	Saliency, gaze or earlier viewers as priors	Content and viewers are shared	On-demand video at 2–10 s horizons	No prior for live or interactive XR
Fetch	Heuristic, buffer-based	Rules map buffer level and view probability to quality	Buffer reflects throughput	Unknown or shifting networks	Hand-tuned; no notion of optimum
	Optimisation	Rate-utility allocation over tiles, cells or LoD	Accurate throughput and view probabilities	Good estimates; 6DoF visibility known	Fragile when estimates are wrong
	Learned (RL)	Policy learned from trace replay	Deployment matches training traces	Stable, well-sampled conditions	Degrades on unseen users and networks
Keep	Popularity cache	LRU or LFU on independent tiles	Requests repeat as whole objects	Large caches; standard HTTP edges	Ignores viewport overlap
	Viewport-aware cache, prefetch	Place tiles or objects by predicted viewing	Viewers' attention overlaps	Popular on-demand content	Gains shown only against popularity baselines
	Shared delivery	Multicast or coded caching of common parts	Synchronised, similar requests	Co-located viewers of one stream	Asynchrony and subpacketisation

[13]. Learned models report large gains mostly over other learned models: a fusion transformer reached an average precision of 0.954 at 1 s against 0.901 for the previous best, but on a random split that placed the same videos and users in training and test [14]. While extrapolation baselines are missing and splits are not disjoint by video and user, nobody can say how much of the reported accuracy is generalisation. A second assumption is that a point estimate is enough. The fetcher needs a distribution to size its margin. Predicting the predictor's own deviation saved 45% of bandwidth at under 0.1% failure [3], and predictors that output several weighted futures exist [15], yet calibration of these probabilities is rarely reported.

1.2 What to fetch: turning probabilities into requests

The fetcher converts a prediction and a throughput estimate into requests: which spatial units, at which quality, for which future segment. It maximises an objective that is assumed rather than measured, a hand-weighted sum of viewport quality, rebuffering and quality switches. Hedging against prediction error is the common core. 360ProbDASH fetched tiles in proportion to their viewing probability and raised viewport PSNR by about 39% [16], and most designs add low-quality margin tiles around the predicted viewport.

The controllers fall into three families. *Heuristic and buffer-based controllers* map buffer level and view probability to quality with fixed rules; BOLA360 extends buffer-based control to tiles with a near-optimality proof and no training, and reports 6–110% QoE gains [17]. *Optimisation-based controllers* solve a rate-utility allocation per segment, often a knapsack over tiles or 3D cells [18, 19]; they are exact for their model and fragile when its inputs are wrong. *Learned controllers*, mostly deep reinforcement learning, learn the policy from trace replay and report 20–30% gains over heuristics [20, 21]. They win on the distribution they were trained on and degrade on unseen viewing patterns and networks [22, 23]. The buffer-based and learned families have never been compared on shared traces under one objective, so their ranking remains open.

Six degrees of freedom change the unit of adaptation from a tile to a 3D cell, an object or a level of detail (LoD), and visibility then depends on position, distance and occlusion as well as direction. ViVo combined these three visibility tests and saved 40% of data on average over commercial 5G at SSIM above 0.99 [24]. No general allocation policy has emerged: the best multi-object strategy depends on the scene and the camera path [25], and the newest public navigation traces come from 15 participants [26].

Live delivery removes both the pre-encoded quality ladder and the cross-user history. YouTube Live delivered 360-degree streams with 4.2–44.2 s camera-to-display delay depending on mode [27]. Two-way volumetric communication breaks the buffer model altogether. Such systems reuse low-latency DASH [28] or WebRTC [29]; a tiled user-adaptive point-cloud system needed 14 Mbps instead of about 300 Mbps in a 33-user study [30]. In these systems adaptation is coupled loosely, if at all, to congestion control (**??**).

1.3 Where to keep it: buffers, caches and shared delivery

Keeping content close buys the right to be wrong cheaply. The client buffer holds bets already placed; an edge cache holds what other viewers, or this viewer's next head turn, may need; shared delivery sends one copy to many. All three are worth more when viewers' attention overlaps.

Popularity caches (LRU, LFU) treat tiles as independent objects, need no prediction and are what deployed HTTP caches run. *Viewport-aware caches and prefetchers* place tiles by predicted or observed cross-user viewing. Placing tiles by viewport similarity raised the edge hit ratio by at least 40% over LFU and 17% over LRU [31], and OpCASH covered more than 95% of the viewport from cache after 24 views while fetching 85% less data from the origin [32]. Independent groups replicate the advantage [33–35], but always against popularity baselines rather than an oracle such as Belady's. Prefetching extends beyond video: world-scale AR model retrieval missed 20% fewer deadlines with ten times fewer wasted bytes [36], and volumetric prefetchers have appeared [37]. *Shared delivery* exploits overlap on the air rather than in storage. Multicast of similar viewports saved over 50% [38], and coded caching promises gains that grow with the number of users, although subpacketisation and the need for synchronised requests may erase them [39]. Joint designs that also transcode or render at the edge report lower latency in simulation [40, 41], a result that holds almost by construction, because the joint problem contains the decoupled one.

Three questions are unresolved. The first is granularity: fixed tiles, variable tiles that rarely recur [32] and similarity caching over continuous orientations [42] have never been compared. The second is the interaction with the fetcher: cache hits distort the throughput estimates that DASH clients rely on [43], yet most caching designs assume an ideal client. The third is the value of learning, which cannot be settled while XRFlux remains the only shared benchmark [44]. Beneath all three lies the assumption that hit ratio stands for experience; no XR caching scheme has been evaluated with viewers. 6DoF content, reusable across viewpoints, users and levels of detail, is the natural next target and remains almost unstudied.

1.4 One loop: horizon, misprediction and realistic savings

The three decisions share one variable, the time between commitment and display. A deep buffer absorbs throughput dips but commits bits on predictions seconds ahead; a shallow buffer needs only short predictions but stalls when throughput drops. Buffer length is therefore prediction horizon, and with accuracy falling from 90% at 0.2 s to 35% at 3 s [1], a deep buffer fills with tiles the user may never see. Short segments ease the tension: users drift by more than 1.3 radians within 5 s, so segments of at most 2–3 s are needed [45], and upgrading buffered tiles lets the fetcher revise a bet as the prediction sharpens. An edge cache shortens the effective horizon by moving the commitment closer to display, so caching and prediction are complements, not substitutes. The same horizon reappears in remote rendering, where pose prediction hides the network round trip rather than throughput variation (**??**).

A wrong bet is paid in one of two currencies. A thin margin lets the user turn into low-quality or blank tiles; a wide margin spends the saving. Tiling adds a fixed cost on top: a median coding overhead of 7.3–10.4% for 1-s tiles [1], up to 37.3% for an 8×5 grid [46] (**??**), and a rise in Flare's phone CPU load from 18% to 58% [1].

The net saving falls as the evaluation becomes real. Perfect knowledge of future motion bounds it at 80% [3]. A scripted viewport, compared with all tiles at high quality, gave up to 72% [47]; HEVC tiles saved 65% with a static viewport and over 40% with recorded head motion [46]; prediction with redundancy saved 45% [3]; Flare on phones over commercial LTE, with four users and three videos, saved 26–35% [1]. The gradient recurs in 6DoF. Visibility-aware delivery of the same content saved 40% for headset users but 10% for phone users [24], because people move differently on each device, while gaze-driven volumetric delivery on HoloLens 2 reported up to 67% [48]. It is established that viewport-adaptive delivery saves a substantial share of bytes. How large that share is on wireless headsets with real viewers, and whether viewers notice it, is not. Most evidence comes from one on-demand client on an emulated link. Only two systems used commercial cellular traces [24, 49], mmWave variability already demands mechanisms such as band switching [50], and mainstream quality models failed on a database of 378 viewport-adaptive videos [51]. Several headsets behind one bottleneck, the normal case in a home or a venue, have reached users in no study.

Open problems

P1. Do predictor rankings survive in closed loop? Persistence nearly matches learned predictors offline, and accuracy is rarely tied to streaming outcome. A tiled player running extrapolators and learned predictors under the same controller, over real bandwidth, with video- and user-disjoint splits and a within-subjects rating study, would show whether accuracy gains reach viewers.

P2. Can calibrated uncertainty replace fixed margins? Controllers need view probabilities but receive point estimates. Distribution-free calibration wrapped around any predictor, with margins chosen for a target miss rate, would show how much bandwidth honest uncertainty saves.

P3. What objective should the fetcher maximise? Weights of quality, stalls and switches are chosen by hand, and quality models fail on viewport-adaptive streams. Fitting the objective to user ratings, or adopting tile extensions of standard models [52, 53], would make controller rankings meaningful.

P4. How should adaptation behave over real wireless access and shared bottlenecks? Paired head-motion and radio traces from Wi-Fi and commercial 5G, replayed and run live with several headsets on one access point, would test whether radio hints and access-point coordination reduce stalls and unfairness.

P5. What should an edge keep for 6DoF content, and do cache gains reach viewers? A level-of-detail cache shared by co-located users, and an HTTP edge serving tiled DASH under popularity, viewport-aware and Belady policies with viewers, would show whether hit-ratio gains are perceptible.

References

[1] Feng Qian, Bo Han, Qingyang Xiao, et al. Flare. In *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, 2018.

[2] Shahryar Afzal, Jiasi Chen, and Kadangode K. Ramakrishnan. Viewing the 360deg Future: Trade-Off Between User Field-of-View Prediction, Network Bandwidth, and Delay. In *Proceedings/Proceedings - International Conference on Computer Communications and Networks*, 2020.

[3] Yanan Bao, Huasen Wu, Tianxiao Zhang, et al. Shooting a moving target: Motion-prediction-based transmission for 360-degree videos. In *IEEE International Conference on Big Data (BigData)*, 2016.

[4] Serhan Gul, Sebastian Bosse, Dimitri Podborski, et al. Kalman Filter-based Head Motion Prediction for Cloud-based Mixed Reality. In *ACM International Conference on Multimedia (ACM MM)*, 2020.

[5] Fang-Yi Chao, Cagri Ozcinar, and Aljosa Smolic. Transformer-based Long-Term Viewport Prediction in 360deg Video: Scanpath is All You Need. In *IEEE International Workshop on Multimedia Signal Processing (MMSP)*, 2021.

[6] Ching-Ling Fan, Jean Lee, Wen-Chih Lo, et al. Fixation Prediction for 360deg Video Streaming in Head-Mounted Virtual Reality. In *nan*, 2017.

[7] Kaiyuan Hu, Haowen Yang, Yili Jin, et al. Understanding User Behavior in Volumetric Video Watching: Dataset, Analysis and Prediction. In *ACM International Conference on Multimedia (ACM MM)*, 2023.

[8] Yixuan Ban, Lan Xie, Zhimin Xu, et al. CUB360: Exploiting Cross-Users Behaviors for Viewport Prediction in 360 Video Adaptive Streaming. In *IEEE International Conference on Multimedia and Expo (ICME)*, 2018.

[9] Lan Xie, Xinggong Zhang, and Zongming Guo. CLS. In *ACM International Conference on Multimedia (ACM MM)*, 2018.

[10] Chenge Li, Weixi Zhang, Yong Liu, et al. Very Long Term Field of View Prediction for 360-Degree Video Streaming. In *IEEE Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2019.

[11] Xianglong Feng, Yao Liu, and Wei Sheng. LiveDeep: Online Viewport Prediction for Live Virtual Reality Streaming Using Lifelong Deep Learning. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2020.

[12] Xianglong Feng, Zeyang Bao, and Sheng Wei. LiveObj: Object Semantics-based Viewport Prediction for Live Mobile Virtual Reality Streaming. *IEEE Transactions on Visualization and Computer Graphics*, 2021.

[13] Gazi Karam Illahi, Ashutosh Vaishnav, Teemu Kamarainen, et al. Learning to Predict Head Pose in Remotely-Rendered Virtual Reality. In *ACM Multimedia Systems Conference (MMSys)*, 2023.

[14] Zhiahao Zhang, Yiwei Chen, Weizhan Zhang, et al. Tile Classification Based Viewport Prediction with Multi-modal Fusion Transformer. In *ACM International Conference on Multimedia (ACM MM)*, 2023.

[15] Quentin Guimard, Lucile Sassatelli, Francesco Marchetti, et al. Deep variational learning for multiple trajectory prediction of 360deg head movements. In *ACM Multimedia Systems Conference (MMSys)*, 2022.

[16] Lan Xie, Zhimin Xu, Yixuan Ban, et al. 360ProbDASH. In *ACM International Conference on Multimedia (ACM MM)*, 2017.

[17] Ali Zeynali, Mohammad Hajiesmaili, and Ramesh K. Sitaraman. BOLA360. In *nan*, 2024.

[18] Jounsup Park, Philip A. Chou, and Jenq-Neng Hwang. Rate-Utility Optimized Streaming of Volumetric Media for Augmented Reality. *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 2019.

[19] Lisha Wang, Chenglin Li, Wenrui Dai, et al. QoE-Driven Adaptive Streaming for Point Clouds. *IEEE Transactions on Multimedia*, 2022.

[20] Yuanxing Zhang, Pengyu Zhao, Kaigui Bian, et al. DRL360: 360-degree Video Streaming with Deep Reinforcement Learning. In *nan*, 2019.

[21] Nuowen Kan, Junni Zou, Chenglin Li, et al. RAPT360: Reinforcement Learning-Based Rate Adaptation for 360-Degree Video Streaming With Adaptive Prediction and Tiling. *IEEE Transactions on Circuits and Systems for Video Technology*, 2021.

[22] Duo Wu, Panlong Wu, Miao Zhang, et al. MANSY: Generalizing Neural Adaptive Immersive Video Streaming With Ensemble and Representation Learning. *IEEE Transactions on Mobile Computing*, 2024.

[23] Lai Wei, Yanting Liu, Fangxin Wang, et al. FewVV: Few-Shot Adaptive Bitrate Volumetric Video Streaming With Prompted Online Adaptation. *IEEE Internet of Things Journal*, 2024.

[24] Bo Han, Yu Liu, and Feng Qian. ViVo. In *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, 2020.

[25] Jeroen van der Hooft, Tim Wauters, Filip De Turck, et al. Towards 6DoF HTTP Adaptive Streaming Through Point Cloud Compression. In *ACM International Conference on Multimedia (ACM MM)*, 2019.

[26] Jeremy Ouellette, Jashanjot Singh Sidhu, and Abdelhak Bentaleb. MazeLab: A Large-Scale Dynamic Volumetric Point Cloud Video Dataset With User Behavior Traces. In *ACM Multimedia Systems Conference (MMSys)*, 2025.

[27] Jeroen van der Hooft, Hadi Amirpour, Maria Torres Vega, et al. A Tutorial on Immersive Video Delivery: From Omnidirectional Video to Holography. *IEEE Communications Surveys & Tutorials*, 2023.

[28] Jack Jansen, Shishir Subramanyam, Romain Bouqueau, et al. A pipeline for multiparty volumetric video conferencing. In *ACM Multimedia Systems Conference (MMSys)*, 2020.

[29] Matthias De Fre, Jeroen van der Hooft, Tim Wauters, et al. Scalable MDC-Based Volumetric Video Delivery for Real-Time One-to-Many WebRTC Conferencing. In *nan*, 2024.

[30] Shishir Subramanyam, Irene Viola, Jack Jansen, et al. Evaluating the Impact of Tiled User-Adaptive Real-Time Point Cloud Streaming on VR Remote Communication. In *ACM International Conference on Multimedia (ACM MM)*, 2022.

[31] Anahita Mahzari, Afshin Taghavi Nasrabadi, Aliehsan Samiei, et al. FoV-Aware Edge Caching for Adaptive 360deg Video Streaming. In *ACM International Conference on Multimedia (ACM MM)*, 2018.

[32] Chamara Madarasingha, Kanchana Thilakarathna, and Albert Y. Zomaya. OpCASH: Optimized Utilization of MEC Cache for 360-Degree Video Streaming with Dynamic Tiling. In *IEEE International Conference on Pervasive Computing and Communications*, 2022.

[33] Niklas Ferdinand Carlsson and Derek L. Eager. Had You Looked Where I'm Looking? Cross-user Similarities in Viewing Behavior for 360-degree Video and Caching Implications. In *ACM/SPEC International Conference on Performance Engineering (ICPE)*, 2020.

[34] Dongbiao He, Jinlei Jiang, Teng Ma, et al. CUBIST: High-Quality 360-Degree Video Streaming Services via Tile-based Edge Caching and FoV-Adaptive Prefetching. In *IEEE International Conference on Web Services (ICWS)*, 2021.

[35] Ziwen Ye, Qing Li, Xiaoteng Ma, et al. VRCT: A Viewport Reconstruction-Based 360deg Video Caching Solution for Tile-Adaptive Streaming. *IEEE Transactions on Broadcasting*, 2023.

[36] Yi-Zhen Tsai, James Luo, Yunshu Wang, et al. The World is Too Big to Download: 3D Model Retrieval for World-Scale Augmented Reality. In *ACM Multimedia Systems Conference (MMSys)*, 2023.

[37] Guanghui Zhang, Shuquan Liu, Mengbai Xiao, et al. PACE: A Multi-Round Cell-Enhanced Prefetching Strategy in Volumetric Video Streaming. *IEEE Transactions on Mobile Computing*, 2025.

[38] Yanan Bao, Tianxiao Zhang, Amit Pande, et al. Motion-Prediction-Based Multicast for 360-Degree Video Transmissions. In *IEEE International Conference on Sensing, Communication, and Networking (SECON)*, 2017.

[39] MohammadJavad Salehi, Kari Hooli, Jari Hulkkonen, et al. Enhancing Next-Generation Extended Reality Applications With Coded Caching. *IEEE Open Journal of the Communications Society*, 2023.

[40] Jianmei Dai, Zhilong Zhang, Shiwen Mao, et al. A View Synthesis-Based 360deg VR Caching System Over MEC-Enabled C-RAN. *IEEE Transactions on Circuits and Systems for Video Technology*, 2019.

[41] Han Xiao, Changqiao Xu, Zichen Feng, et al. A Transcoding-Enabled 360deg VR Video Caching and Delivery Framework for Edge-Enhanced Next-Generation Wireless Networks. *IEEE Journal on Selected Areas in Communications*, 2022.

[42] Anirudh Sabnis, Tareq Si Salem, Giovanni Neglia, et al. GRADES: Gradient Descent for Similarity Caching. In *nan*, 2021.

[43] Anahita Mahzari, Aliehsan Samiei, and Ravi Prakash. CooPEC: Cooperative Prefetching and Edge Caching for Adaptive 360deg Video Streaming. In *IEEE International Symposium on Multimedia (ISM)*, 2020.

[44] Nader Alfares and George Kesidis. Virtual Reality Benchmark for Edge Caching Systems. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2025.

[45] Xavier Corbillon, Gwendal Simon, Alisa Devlic, et al. Viewport-adaptive navigable 360-degree video delivery. In *International Conference on Communications*, 2017.

[46] Mario Graf, Christian Timmerer, and Christopher Mueller. Towards Bandwidth Efficient Adaptive Streaming of Omnidirectional Video over HTTP. In *nan*, 2017.

[47] Mohammad Hosseini and Viswanathan Swaminathan. Adaptive 360 VR Video Streaming: Divide and Conquer. In *IEEE International Symposium on Multimedia (ISM)*, 2016.

[48] Nan Wu, Kaiyan Liu, Ruizhi Cheng, et al. Theia: Gaze-driven and Perception-aware Volumetric Content Delivery for Mixed Reality Headsets. In *International Conference on Mobile Systems, Applications, and Services*, 2024.

[49] Liyang Sun, Fanyi Duanmu, Yong Liu, et al. Multi-path multi-tier 360-degree video streaming in 5G networks. In *ACM Multimedia Systems Conference (MMSys)*, 2018.

[50] Eman Ramadan, Arvind Narayanan, Udhaya Kumar Dayalan, et al. Case for 5G-aware video streaming applications. In *nan*, 2021.

[51] Xuelin Liu, Haoyun Zhang, Jiebin Yan, et al. Quality of Experience of Viewport Adaptive Omnidirectional Video Streaming. In *Proceedings - International Conference on Image Processing*, 2024.

[52] Yuichiro Urata, Masanori Koike, Kazuhisa Yamagishi, et al. Extension of ITU-T P.1203 model to Tile-based Omnidirectional Video Streaming. *Electronic Imaging*, 2021.

[53] Masanori Koike, Yuichiro Urata, Noritsugu Egi, et al. Extension of ITU-T P.1204.3 Model to Tile-Based VR Streaming Services. In *IEEE International Workshop Technical Committee on Communications Quality and Reliability (CQR)*, 2021.